MATH317 · NUMERICAL METHODS

Mock Midterm Solutions

A practice exam in the midterm's format · 60 minutes · 30 marks

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01 Solution to Question 1

#	Answer	Reason
1	F	Divide by the exact value, not the approximate one.
2	T	This is the definition of chopping.
3	F	Maclaurin is about $c = 0$.
4	T	127 for single, 1023 for double.
5	F	It is the smallest positive number with $1 + \varepsilon \neq 1$.
6	T	The interval halves each step, so the bound halves.
7	T	The formula contains $\frac{f(x_n)}{f'(x_n)}$.

02 Solution to Question 2

Coefficients: `2, −5, 0, 7`. Do not forget the zero for the `x` term.

a)	2	−5	0	7
2 |		4	−2	−4
	2	−1	−2	3

P(2) = 3

b)	2	−5	0	7	
1 |		2	−3	−3	
	2	−3	−3 |	4	c0 = 4
		2	−1		
	2	−1 |	−4		c1 = −4
		2			
	2 |	1			c2 = 1
	2				c3 = 2

$P(x) = 2(x-1)^3 + (x-1)^2 - 4(x-1) + 4$

Check: `P(1) = 2 − 5 + 7 = 4`.

03 Solution to Question 3

Negative → s = 1

$37 = (45)_8 = (100101)_2$

$.40625 \times 8 = 3.25 \rightarrow 3$

$.25 \times 8 = 2.0 \rightarrow 2$

$.40625 = (.32)_8 = (.011010)_2$

$37.40625 = (100101.01101)_2 = 1.0010101101 \times 2^5$

$c = 5 + 127 = 132 = (204)_8 = (10000100)_2$

1 | 10000100 | 00101011010000000000000

1100 0010 0001 0101 1010 0000 0000 0000

$= [C215A000]_{16}$

04 Solution to Question 4

a) $u = f(1) = 1 + 1 - 3 = -1$ $v = f(2) = 8 + 2 - 3 = 7$

$u \cdot v < 0 \rightarrow$ f has a root in [1, 2]

$f'(x) = 3x^2 + 1 > 0 \rightarrow$ f is increasing → exactly one root

b) $n = 0$: $c = \frac{1 + 2}{2} = 1.5$ $w = f(1.5) = 1.875$

$u \cdot w < 0 \rightarrow b = c \rightarrow [1, 1.5]$

$n = 1$: $c = \frac{1 + 1.5}{2} = 1.25$ $w = f(1.25) = 0.2031$

$u \cdot w < 0 \rightarrow b = c \rightarrow [1, 1.25]$

$n = 2$: $c = \frac{1 + 1.25}{2} = 1.125$ $w = f(1.125) = -0.4512$

$u \cdot w > 0 \rightarrow a = c \rightarrow [1.125, 1.25]$

n	c_n	$f(c_n)$	Error
0	1.5	1.875	0.5
1	1.25	0.2031	0.25
2	1.125	−0.4512	0.125

We stop because the iteration count reached `n_max = 2`.

05 Solution to Question 5

$$f'(x) = 3x^2 + 1$$

$$f(1) = -1 \qquad f'(1) = 4$$

$$x_1 = 1 - \frac{-1}{4} = 1.25$$

$$f(1.25) = 0.2031 \qquad f'(1.25) = 5.6875$$

$$x_2 = 1.25 - \frac{0.2031}{5.6875}$$

$$\mathbf{x_2 = 1.2143}$$

Compare: after two steps Newton reached `1.2143`, while bisection after three steps only says the root is between `1.125` and `1.25`.

06 Grade yourself

If you missed	Review
Q1	The definitions in every chapter
Q2	Chapter 1.1 · Chapter 1.2
Q3	Chapter 2
Q4	Chapter 3.1
Q5	Chapter 3.2